

XUAN LUO

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EDUCATION

University of Washington <i>PhD Student in Computer Science & Engineering</i> Advisors: Steven M. Seitz, Jason Lawrence, and Ricardo Martin-Brualla.	Sept. 2015 – 2022 Seattle, WA
Shanghai Jiao Tong University <i>BS in Computer Science & Technology</i> ACM Honors Class (one of the top gifted CS programs in China).	Sept. 2011 – June 2015 Shanghai, China

AREAS OF EXPERTISE

Long Context Modeling, Efficient Transformer, 3D Vision, Image Synthesis, Computational Photography

PROFESSIONAL EXPERIENCE

Google <i>Research Scientist</i> Project Beam — the future of communications.	2022 – Present Seattle, WA
• Drove the core development of a real-time 3D communication system, successfully demonstrating the importance and feasibility of temporal novel view synthesis systems for next-generation products, setting a new technical direction for the team. • Integrating 3D geometry, enabled transformers to also achieve real-time, 4K novel view synthesis. • Designed long-context modeling mechanisms, balancing long-term memorization with real-time constraints. • Developed sparse attention algorithms to enable high-resolution novel view synthesis at real-time speeds. • Co-authored <i>Quark</i>, a real-time, high-resolution, and generalizable neural view synthesis framework (Best Paper, SIGGRAPH Asia 2024). • Developed diffusion-based generative models for photorealistic 3D avatars.	
University of Washington <i>Graduate Student Researcher</i> Advisors: Steven M. Seitz, Jason Lawrence, and Ricardo Martin-Brualla	2015 – 2022 Seattle, WA
• Restored what famous historical figures would look like if rephotographed with modern cameras. • Collected a large-scale rectified historical stereo dataset and visualized historical scenes in 3D. • Designed an inexpensive glass-free DIY 3D display with a tablet and a plastic sheet folded into a cone.	
Facebook <i>Research Intern</i> Mentor: Johannes Kopf. Collaborators: Jia-bin Huang, Kevin Matzen, and Richard Szeliski	June 2019 – March 2020 Seattle, WA
• Estimated geometrically consistent depth from monocular videos, enabling video effects to a whole new level.	
Disney Research Zurich <i>Research Intern</i> Collaborators: Thabo Beeler, Derek Bradley, Matthias Niessner, and Paulo Gotardo.	Summer 2017 Zurich, Switzerland
• Worked on facial motion capture.	
Google Daydream <i>Software Engineering Intern</i> Mentor: Jason Lawrence	Summer 2016 Seattle, WA
National University of Singapore <i>Visiting Scholar</i> Advisor: Shuicheng Yan	Aug. 2014 – Feb. 2015 Singapore

PUBLICATIONS

<i>Online Neural Space Time Memory for Dynamic Novel View Synthesis</i> Baback Elmieh, Lynn Tsai, Zeman Li, Srinivas Kaza, Tiancheng Sun, Gabor Csapo, Ali Behrouz, Yuan Deng, Stephen Lombardi, Steven M. Seitz, Xuan Luo	2026
<i>Scaling Transformer-Based Novel View Synthesis with Models Token Disentanglement and Synthetic Data</i> Nithin Gopalakrishnan Nair, Srinivas Kaza, Xuan Luo , Vishal M. Patel, Stephen Lombardi, Jungyeon Park	ICCV 2025

<i>Learning Efficient Fuse-and-Refine for Feed-Forward 3D Gaussian Splatting</i>	NeurIPS 2025
Yiming Wang, Lucy Chai, Xuan Luo , Michael Niemeyer, Manuel Lagunas, Stephen Lombardi, Siyu Tang, Tiancheng Sun	
<i>LVT: Large-Scale Scene Reconstruction via Local View Transformers</i>	SIGGRAPH Asia 2025
Tooba Imtiaz*, Lucy Chai*, Kathryn Heal, Xuan Luo , Jungyeon Park, Jennifer Dy, John Flynn	
<i>Quark: Real-time, High-resolution, and General Neural View Synthesis</i>	SIGGRAPH Asia 2024
<i>Best Paper Award</i>	
John Flynn*, Michael Broxton*, Lukas Murmann*, Lucy Chai, Matthew DuVall, Clément Godard, Kathryn Heal, Srinivas Kaza, Stephen Lombardi, Xuan Luo , Supreeth Achar, Kira Prabhu, Tiancheng Sun, Lynn Tsai, Ryan Overbeck	
<i>StyleSDF: High-Resolution 3D-Consistent Image and Geometry Generation</i>	CVPR 2022
Roy Or-El, Xuan Luo , Mengyi Shan, Eli Shechtman, Jeong Joon Park, Ira Kemelmacher-Shlizerman	
<i>Time-Travel Rephotography SIGGRAPH Asia</i>	TOG 2021
Xuan Luo , Cecilia Zhang, Paul Yoo, Ricardo Ricardo Martin-Brualla, Jason Lawrence, and Steven M. Seitz	
<i>Consistent Video Depth Estimation SIGGRAPH</i>	TOG 2020
Xuan Luo , Jia-Bin Huang, Richard Szeliski, Kevin Matzen, and Johannes Kopf	
<i>KeystoneDepth: History in 3D International Virtual Conference on 3D Vision</i>	3DV 2020
Xuan Luo , Yanmeng Kong, Jason Lawrence, Ricardo Martin-Brualla, and Steven M. Seitz	
<i>Slow Glass: Visualizing History in 3D Fourth Workshop on Computer Vision for AR/VR</i>	CVPR-W 2020
Xuan Luo , Yanmeng Kong, Jason Lawrence, Ricardo Martin-Brualla, and Steven M. Seitz	
<i>Pepper's Cone: An Inexpensive Do-It-Yourself 3D Display</i>	UIST 2017
Xuan Luo , Jason Lawrence, and Steven M. Seitz	
<i>Purine: A Graph-based Deep Learning Framework</i>	ICLR 2015
Min Lin, Shuo Li, Xuan Luo , and Shuicheng Yan	

HONORS AND AWARDS

Best Paper Award, SIGGRAPH Asia	2024
Selected EECS Rising Star by EECS at UC Berkeley	Nov. 2020
Anne Dinning - Michael Wolf Endowed Regental Fellowship, UW	2015 – 2016
Distinguished Undergraduate Scholarship, SJTU	2015
Shanghai Outstanding Graduate, Shanghai	2015
National Scholarship, China <i>Highest scholarship in China</i>	2013
2012 University Physics Competition, Silver Medal, USA	2012

MEDIA & PRESS

<i>Time-Travel Rephotography</i>	2020
<u>Two Minute Papers</u> , <u>GIZMOD0</u> , <u>Hack a Day</u> , <u>QubitAI</u> , <u>Guokr</u> , <u>Tencent</u> , <u>marktechpost.com</u> , <u>TechTheLead</u> , <u>Tech Times</u> , <u>The Science Times</u> , <u>Review Geek</u> , <u>Tech Explore</u> , <u>msnice.net</u> , <u>kknews.cc</u>	
<i>Consistent Video Depth</i>	2020
<u>Two Minute Papers</u> , <u>QubitAI</u> , <u>Synced</u> , <u>Medium</u> , <u>Bo Yu AI</u>	
<i>Pepper's Cone</i>	2018
<u>"Demo Hour" of ACM Interactions Magazine</u> , <u>iProgrammer</u> , <u>Hack a Day</u> , <u>Hacker News</u>	

PROFESSIONAL SERVICE

WiGRAPH: Women in Graphics Research (wigraph.org)	Fall 2020 – 2025
Rising Stars Coordinator	
Poster Juror	
• The ACM Special Interest Group on Computer Graphics (SIGGRAPH)	2025